Assignment 3a Reflection - Ben Branta

1. Describe a mistake you made or error message you got while solving the problems, and how you will recognize in the future that you made the mistake again.

Answer: In advanced fizzbuzz I did not have a validation check for if n = 0. The test case fizzbuzz__adv(0) created an infinite loop from my valuation(0, 3) call. In valuation I don't have a check if n = 0. If n = 0 then my while loop permanently evaluates to true causing the infinite loop.

This was recognizable because when I test run my script it never completes. Looking ahead I think well defined test cases catch these issues. As well as looking for those conditions which can cause issues.

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2. What makes a good test case for determining if a function works correctly? How do you know that your functions perform correctly?

Answer: A good test case to me would check that the outputs are correct, corner cases are handled, and inputs are validated. It should check and make sure the function handles errors gracefully. I think a well defined function with a single purpose would perform correctly if it passes the test cases.

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3. Did you make any changes to your mortgage calculator other than package it into a function?

Answer: In assignment 2 I prepared it as a function with control flow so it was an easy conversion. I did add helper functions for input validation and calculating the monthly interest. The input validation was a nice way to improve my mortgage calculator function because it adds many lines of code that detract from the actual purpose of the function. The calculate monthly interest function was a nice way to turn a math equation into a meaningful action that is readable. Equations are up to the reader to interpret so giving it a function name helps a reader understand its purpose. I did not add anything beyond those two because the rest of it was relatively easy to understand.

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4. Does the returning of None, None in the infinite loop sufficiently communicate to the caller that there was an issue with the inputs? What might be done differently?

Answer: No it does not sufficiently communicate there was an issue. In assignment 2 I used try/except and raised exceptions with text descriptions of the issue. Here I returned None, None but triggered a print statement explaining the issue as well. Adding a text description or a failure code would help.

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5. Describe one thing that you learned while working on this lesson that stood out as useful or interesting.

Answer: I struggle the most with input validation and looking for error cases. In this lesson I created an infinite loop via a helper function. I found that by working with a coding agent and asking for failure cases of my function that it helps prevent creating errors that I didn't even know to think of. It is much more capable of processing these requests and each time it finds a mistake I know to look for it next time.